

The *domain.py* consists of the domain model classes, *data.py* consists of *UserRepositoryImpl* and *app.py* consists of the rest of the code. Since we have seen the code already, let us move on to the configuration files.

The first one is the file named *requirements.txt*. It declares the external dependencies for the Python system to download and install. We need to populate it with every external Python module that we use in the *FindService*. As you know, we used MySQL connector, Flask and Waitress modules. Hence the following is the content of the *requirements.txt*.

```
Flask==2.1.1
Flask_RESTful
mysql-connector-python
waitress
```

The second step is to declare the manifestation for dockerisation in *Dockerfile*. Here it is:

```
FROM python:3.8-slim-buster

WORKDIR /ums
ADD ums /ums
ADD requirements.txt requirements.txt
RUN pip3 install -r requirements.txt

EXPOSE 8080
ENTRYPOINT ["python"]
CMD ["/ums/app.py"]
```

In summary, we used Python 3.8 as the baseline and moved our code from the *ums* folder to a corresponding folder in the Docker container, besides moving the *requirements.txt*. Then we instructed the container to run the *pip3 install* command. And, finally, we exposed port 8080 (since the *waitress* was running on that port).

In order to run the service, we instructed the container to use the following command:

```
python /ums/app.py
```

Once the *Dockerfile* is ready, run the following command from within the *ums-find-service* folder to create the Dockerised image:

```
docker build -t glarimy/ums-find-service
```

It creates the Docker image, which can be found using the following command:

```
docker images
```

Try pushing the image to the Docker Hub as appropriate. You may log in to Docker as well.

```
docker login
docker push glarimy/ums-find-service
```

And the last step is to build the manifest for the Kubernetes deployment.

We have already covered the way to set up the Kubernetes cluster, and deploy and use services, in the previous part. I am assuming that we still have the manifest file we used in the previous part for deploying the *AddService*, MySQL, Kafka and Zookeeper. We only need to add the following entries into the *kube-find-deployment.yml* file:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ums-find-service
  labels:
    app: ums-find-service
spec:
  replicas: 3
  selector:
    matchLabels:
      app: ums-find-service
  template:
    metadata:
      labels:
        app: ums-find-service
    spec:
      containers:
        - name: ums-find-service
          image: glarimy/ums-find-service
          ports:
            - containerPort: 8080
---
apiVersion: v1
kind: Service
metadata:
  name: ums-find-service
  labels:
    name: ums-find-service
spec:
  type: LoadBalancer
  ports:
    - port: 8080
  selector:
    app: ums-find-service
```

The first part of the above manifest declares a